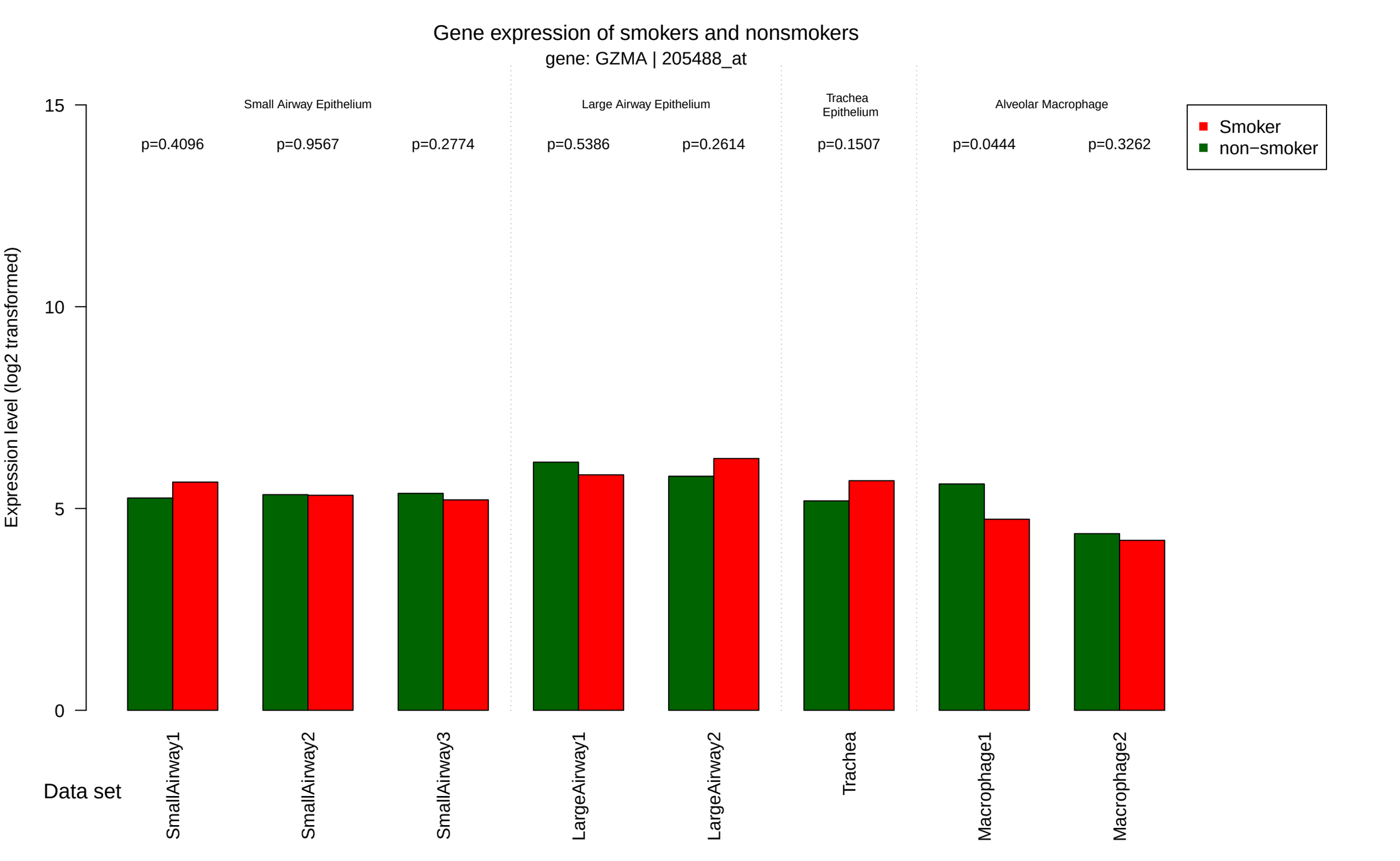
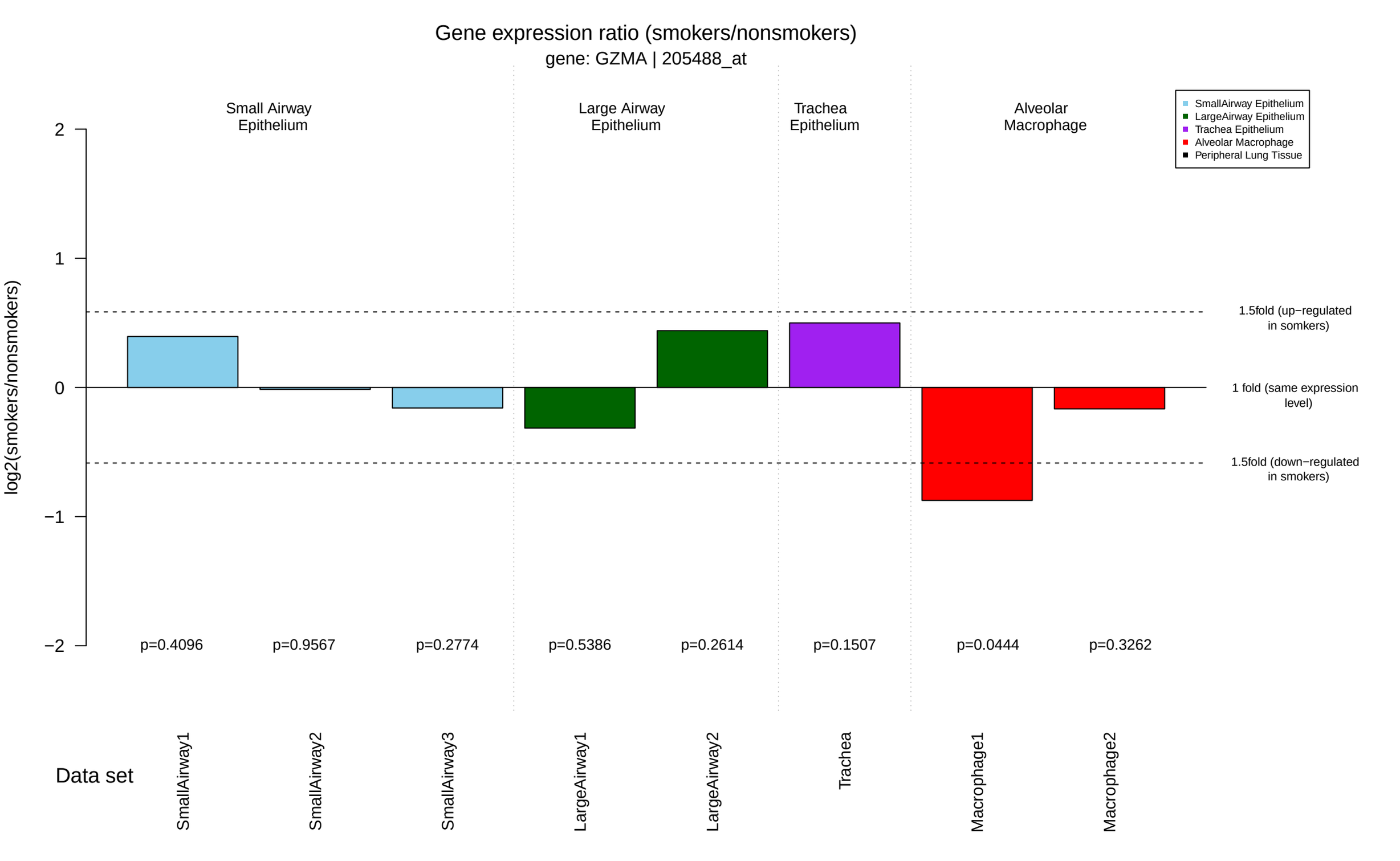
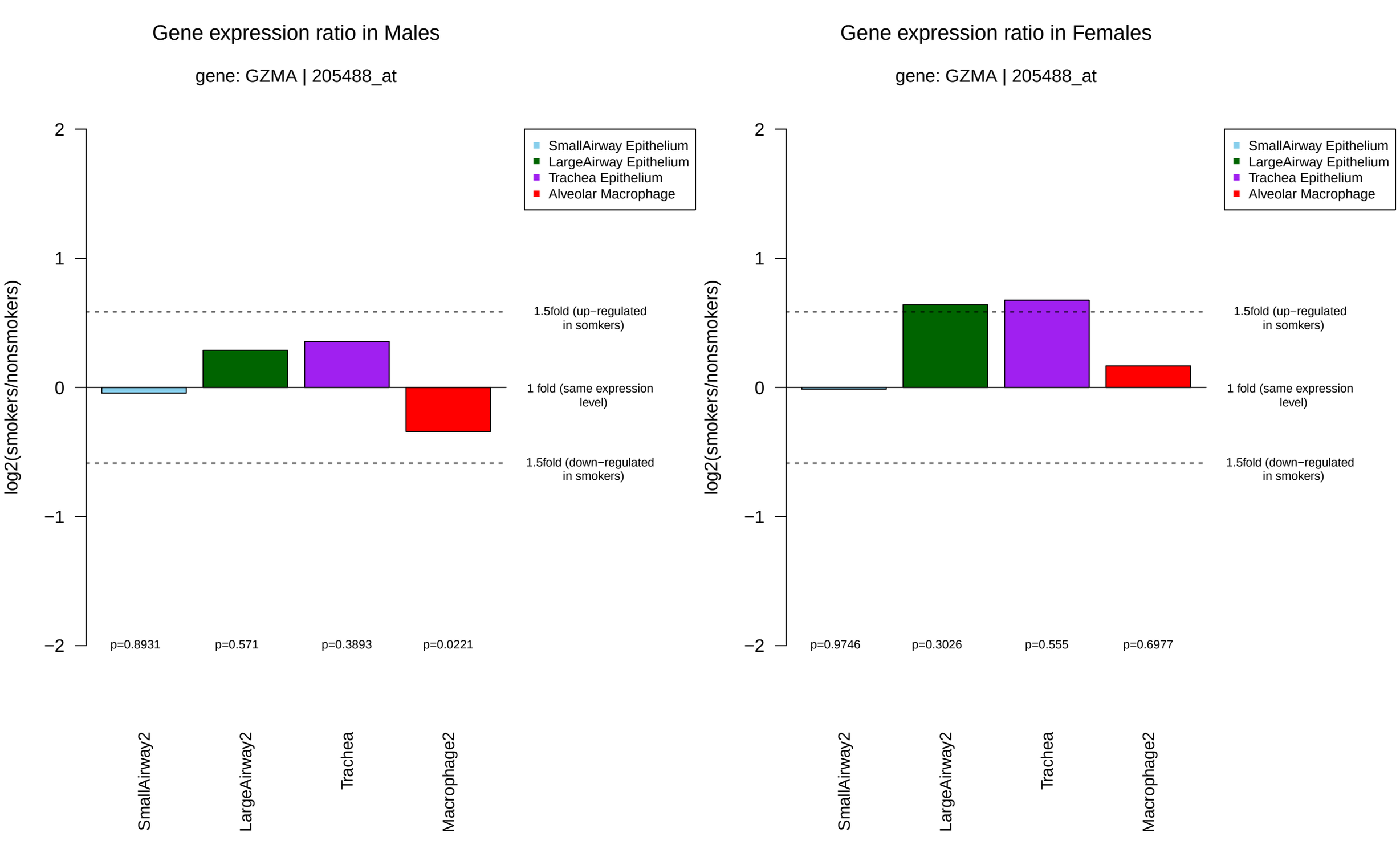
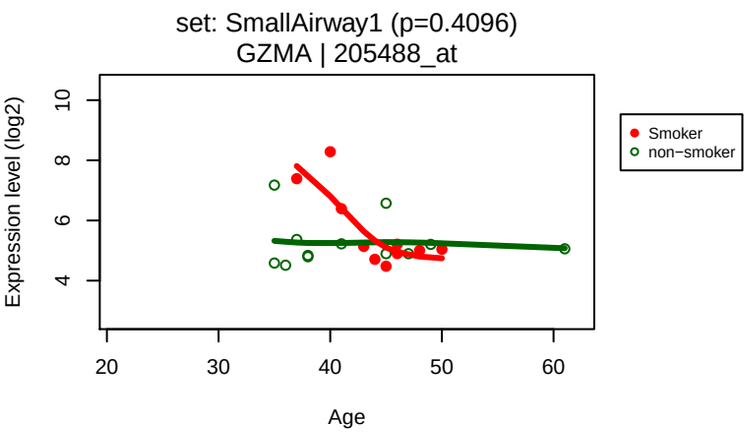
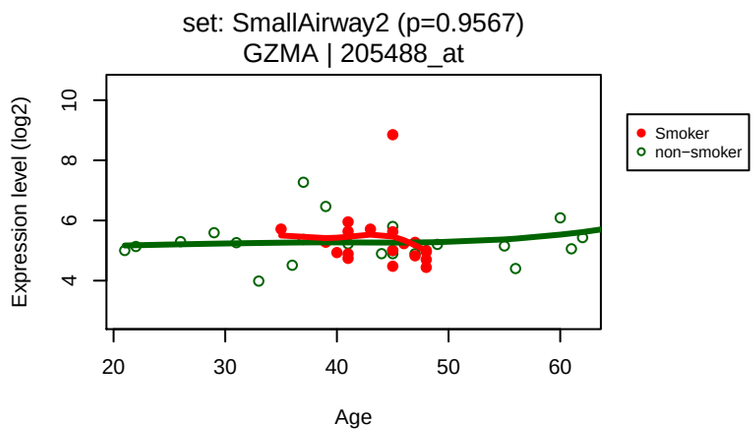
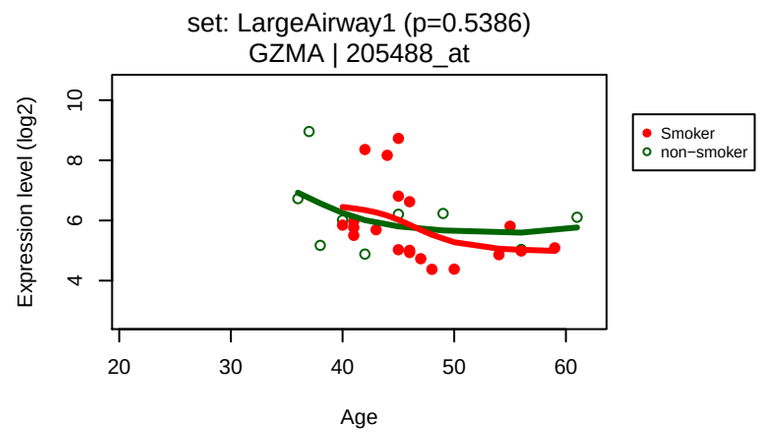

Gene expression of smokers and nonsmokers
gene: GZMA | 205488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4096
p=0.9567
p=0.2774
p=0.5386
p=0.2614
p=0.1507
p=0.0444
p=0.3262
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

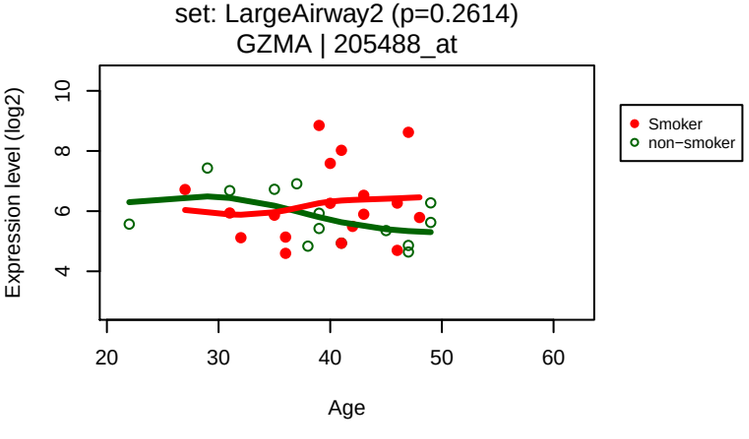

Gene expression ratio (smokers/nonsmokers)
gene: GZMA | 205488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4096
p=0.9567
p=0.2774
p=0.5386
p=0.2614
p=0.1507
p=0.0444
p=0.3262
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

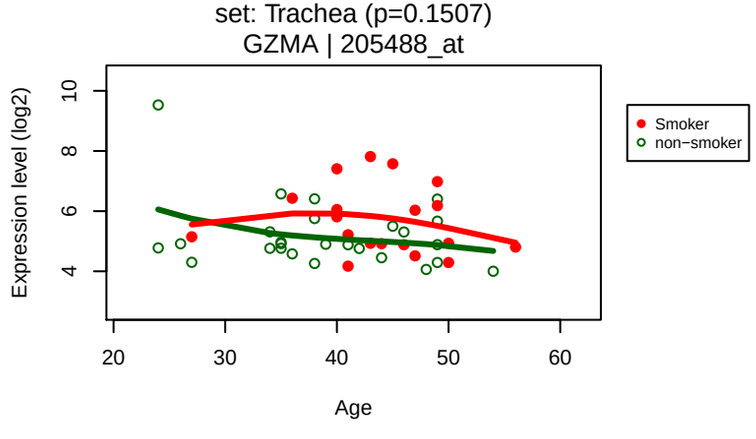

Gene expression ratio in Males
gene: GZMA | 205488_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8931
p=0.571
p=0.3893
p=0.0221
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GZMA | 205488_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9746
p=0.3026
p=0.555
p=0.6977
SmallAirway2
LargeAirway2
Trachea
Macrophage2

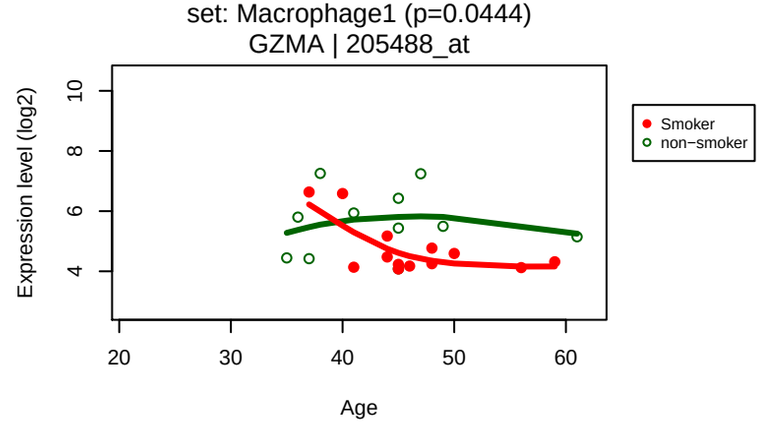

set: SmallAirway1 (p=0.4096)
GZMA | 205488_at
Expression level (log2)
Age
Smoker
non−smoker

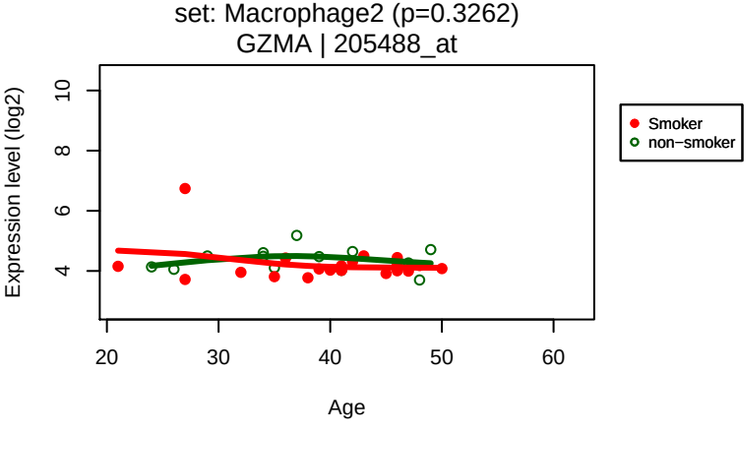

set: SmallAirway2 (p=0.9567)
GZMA | 205488_at
Expression level (log2)
Age
Smoker
non−smoker

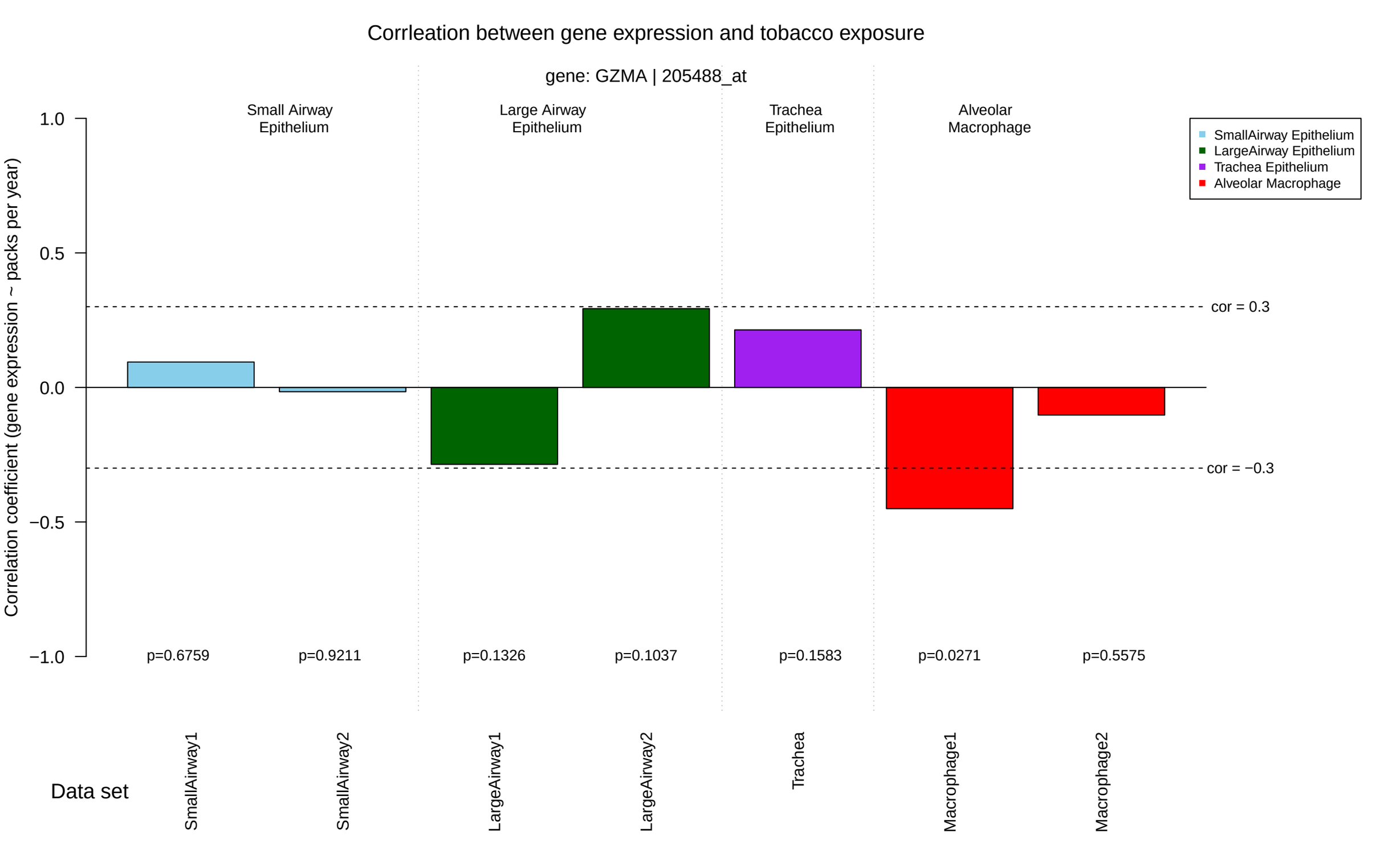

set: LargeAirway1 (p=0.5386)
GZMA | 205488_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2614)
GZMA | 205488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1507)
GZMA | 205488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0444)
GZMA | 205488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3262)
GZMA | 205488_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GZMA | 205488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6759
p=0.9211
p=0.1326
p=0.1037
p=0.1583
p=0.0271
p=0.5575
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2